

Sachin Koli

Solution Architect — Cloud Data & AI Platforms

Google Cloud Certified — Professional Data Engineer

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SUMMARY

Google Cloud Certified Professional Data Engineer with **18 years** building enterprise data platforms, now architecting **real-time streaming and batch data infrastructure on Google Cloud** for a UK tier-1 retail bank — Confluent Kafka, Cloud Composer, dbt, BigQuery and Terraform, spanning legacy on-premise and cloud environments.

Previously architected **digital cards and payments** at the same bank, migrating card servicing to GCP. The self-serve fraud journey I designed won at the **Banking Tech Awards 2024** and the **Card & Payments Awards 2025**.

Across twelve prior years in retail, telecom and banking data: a **Teradata** → **BigQuery** lake migration, and a customer preference API on GCP that **cut \$600,000 a year** in operating cost. Alongside all of it I **ship applied-AI tooling in the open** — a terminal coding agent published to npm, and the Rust beneath it: ONNX inference, an ANN index written from scratch, hybrid dense/lexical retrieval.

EXPERIENCE

Tata Consultancy Services

Sep 2021 – Present

Solution Architect · Pune, India

Client: Lloyds Banking Group, United Kingdom

Data Platform & Real-Time Analytics — Jul 2025 – Present

- Architected the bank's **end-to-end batch and real-time streaming data platform** on Google Cloud, supporting high-throughput analytics, operational reporting and machine learning workloads.
- Engineered high-availability, event-driven ingestion pipelines on **Confluent Kafka** with Python, delivering low-latency data for operational reporting and instant business insight.
- Architected hybrid batch orchestration across **Cloud Composer (Apache Airflow)** and **IBM Tivoli Workload Scheduler**, managing complex dependency graphs that span legacy on-premise and cloud environments.
- Designed optimised **BigQuery** warehouse schemas and modular **dbt** transformation pipelines with enforced data quality and lineage; built automated testing, validation and monitoring frameworks so downstream analytics fail loudly rather than silently.
- Standardised infrastructure with **Terraform** — Composer environments, BigQuery datasets, Kafka connectors and IAM roles — and integrated it into **CI/CD**, cutting configuration drift and deployment cycle time.
- Implemented BigQuery and GCP **cost optimisation and governance**, reducing compute overhead while improving query performance and data security.
- Partnered with leadership, security and business units to turn requirements into production-ready architectural blueprints.

Digital Cards Modernisation — 2022 – Jul 2025

- Led end-to-end modernisation of **Digital Cards** — Add to Wallet, card controls, View PIN / View Card — migrating from on-premise infrastructure to **GCP**.
- Architected secure, scalable **microservices and cross-cloud API patterns** meeting banking compliance and performance requirements.
- **Award-winning self-serve fraud journey** — recognised at the Banking Tech Awards 2024 and the Card & Payments Awards 2025 — moving fraud handling from an assisted call-centre process to a journey the customer completes.
- Designed service discovery, API gateways, containerisation, load balancing and secure inter-service communication across a hybrid estate.

Sears IT and Management Services India

Feb 2018 – Sep 2021

Technical Specialist → Architect · Pune, India

Client: Sears Holdings Corporation, United States — Targeted Interactions / TI Mart

- Built the **Email Preference Center API** on GCP App Engine, Pub/Sub, DataFlow and BigQuery, serving millions of customers — **reduced annual cost by \$600,000**.
- **Migrated DataMarts from Teradata to BigQuery**, on schedule and without significant incident.
- Built an **API management layer** decoupling applications from cloud complexity across a multicloud estate — reused as the blueprint by other teams.
- Replaced self-hosted tooling with cloud-native across five stacks: Teradata → BigQuery · Unix → Compute Engine · Control-M → Airflow · shell/ETL → Python · GitLab Enterprise → Cloud Source Repositories.
- Designed shared datamarts (CDW, NSegment, Campaign) in BigQuery; led enterprise migration from **IBM Unica to Adobe Campaign** and Telluride, coordinating vendors (Adobe, BounceX) and internal teams.
- Delivered Tableau and Data Studio reporting; ran continuous GCP cost optimisation.

Client: Saudi Telecom Company, Saudi Arabia — STC EDW (CLDM7 / CLDM12) · Onsite IT Delivery Lead · Feb 2015 – Feb 2018

- Led data modelling for the enterprise warehouse covering customers, accounts, subscriptions, offers and products across all channels and business units.
- Extended the model by **snowflaking dimensions** — demographics, account status, subscription status.
- Built an **automation tool** propagating a new key through every dimension table, Teradata utility script (FastLoad, MultiLoad, FastExport, BTEQ) and Ab-Initio object automatically.
- Ran a big-data POC on **Apache Hive and HDFS** for CDR ingestion.

Client: Barclays, United Kingdom — Emerging Markets DW & One Africa Program · Team Lead · Nov 2010 – Feb 2015

- Built staging from core banking (FLEX), ATM/POS (SPARROW) and online transaction sources, then the DataMarts above them.
- **One Africa:** unified emerging-market data across African countries into a single source of truth, letting one DataMart span new countries with no rework.
- Designed the **logical and physical data models**; built reusable automation — TWS object-creation tool, Teradata acquisition code generator, Unix deployment tooling.
- Implemented archiving and purging against the active warehouse, tiering history to **Greenplum**.

Client: Telstra Corporation, Australia — TR2.1 / TR2.2 EDW Data-In

- Built **DataStage 7.5** jobs, Teradata utility scripts and Perl/Unix tooling for the retail customer warehouse — registrations, demographics, segmentation, billed and unbilled call records.
- Built **Informatica** mappings and workflows migrating Oracle to Teradata for an enterprise data warehouse.

OPEN SOURCE — APPLIED AI & AGENT TOOLING

Built in my own time and shipped in public — github.com/kolisachint

- **hoocode** — deterministic terminal coding agent, **published to npm**. TypeScript monorepo of four packages: CLI, agent runtime with tool calling, a **unified LLM API across 25+ providers**, and a differential-rendering TUI. Permission gate on every edit and command, four scoped modes, MCP servers and subagents. ~470 commits since May 2026, extending a fork of the MIT-licensed *pi-mono*.
- **embeddingsearchtools** (Rust) — embedding search engine. MiniLM via **ONNX Runtime**, int8 weights bundled into the binary; exact and **HNSW approximate-nearest-neighbour** indexes written from scratch; **BM25 fusion for hybrid retrieval**; mmap-friendly store behind a library API, CLI and daemon.
- **voicetools** (Rust) — offline speech-to-text. Mic capture through voice-activity detection into **Parakeet-TDT on ONNX Runtime** (25 languages, int8), whisper.cpp fallback.
- **webtools** (Rust) — token-efficient fetch and search for LLM agents; reference-style URL preservation collapses links to single-token markers under a total token budget.

TECHNICAL

Cloud	Google Cloud Platform — BigQuery, Cloud Composer, DataFlow, Pub/Sub, App Engine, Cloud Functions, DataProc, Cloud Storage, Compute Engine, Memorystore, Cloud SQL, VPC, IAM · AWS (S3)
Streaming	Confluent Kafka, Pub/Sub, Apache Beam, event-driven architecture
Orchestration	Cloud Composer / Apache Airflow, IBM Tivoli Workload Scheduler, Control-M
Transformation	dbt, Python, SQL, Apache Spark, Informatica, IBM DataStage, Ab-Initio
Infrastructure	Terraform, Git, CI/CD, containerisation
Architecture	Microservices, API gateways, service discovery, load balancing, multicloud and hybrid patterns, data governance, cost optimisation
Databases	BigQuery, Teradata, Oracle, Greenplum
BI	Tableau, Looker Studio, Cognos
Domains	Banking, cards & payments · Retail · Telecom

EDUCATION & CERTIFICATIONS

BE, Computer Science and Engineering — Walchand College of Engineering, Sangli (Shivaji University), 2003–2007

- **Google Cloud Certified — Professional Data Engineer**
- Google Cloud Professional Cloud Architect — *in progress*
- Data Engineering, Big Data and Machine Learning on GCP — Coursera specialisation
- Teradata Basics V2R5 · Teradata SQL V2R5 · Introduction to Oracle9i: SQL